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EAI421- APPLICATION OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PROJECT TITLE:

"Smart Soil & Crop Recommendation System using AI"

Submitted to

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Application of AI&ML

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CERTIFICATE

This is Certified that the Project work entitled **"Smart Soil & Crop Recommendation System using AI"** is a bonafide work carried out by **Varun K R** bearing the university seat number **AMB1270** in partial fulfillment for the award of Bachelor of Science (HONS) Agriculture Degree in University of Agricultural sciences, College of Agriculture, GKVK, Bengaluru during the year 2025-26. It is certified that all corrections/suggestions indicated for the internal assessment have been incorporated in the Project report. The Project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the VI Semester Bachelor of Science (HONS) Agriculture Degree.

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ABSTRACT

Agriculture plays a vital role in ensuring food security and sustainable development. With increasing challenges such as soil degradation, climate variability, and improper fertilizer usage, there is a growing need for intelligent systems that assist farmers in making data-driven decisions. This project presents a Soil Analysis and Crop Recommendation System that utilizes machine learning techniques to analyze soil parameters and recommend suitable crops and fertilizers.

The system takes inputs such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH level, rainfall, and soil type. Based on these parameters, a trained machine learning model predicts either the optimal fertilizer ratio or the most suitable crop. Additionally, the system provides soil health analysis, farming guidelines, and crop-specific recommendations.

The application is developed using Streamlit for an interactive user interface and integrates trained models using Joblib. The system enhances decision-making for farmers, promotes efficient resource utilization, and supports precision agriculture practices.

Keywords: Machine Learning, Soil Analysis, Crop Recommendation, Fertilizer Prediction, Precision Agriculture, NPK, Streamlit

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1.INTRODUCTION

Agriculture is a fundamental sector that supports food security, employment, and economic growth, especially in developing countries like India. The productivity of crops largely depends on soil quality, nutrient availability, and environmental conditions. However, traditional farming practices often rely on experience-based decisions rather than scientific analysis, which can lead to inefficient use of fertilizers, poor crop yield, and soil degradation.

Soil nutrients such as Nitrogen (N), Phosphorus (P), and Potassium (K) are essential for plant growth and development. An imbalance in these nutrients can significantly affect crop productivity. Additionally, environmental factors such as temperature, humidity, rainfall, and soil pH play a crucial role in determining crop suitability. Farmers often face challenges in accurately assessing these parameters and selecting the right crop and fertilizer combination.

With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), it is now possible to analyze large datasets and extract meaningful insights for decision-making. Machine learning models can identify patterns in soil and environmental data and provide accurate recommendations for crop selection and fertilizer usage.

This project focuses on the development of a **Smart Soil & Crop Recommendation System using Artificial Intelligence**. The system takes various soil and environmental inputs and predicts suitable crops or fertilizer recommendations. It also provides soil analysis and smart farming guidelines, making it a comprehensive decision-support tool.

The proposed system aims to enhance agricultural productivity, optimize resource utilization, and promote sustainable farming practices by integrating modern AI techniques into traditional agriculture.

2.OBJECTIVE

Main Objective

To design and develop a smart AI-based system that recommends suitable crops and fertilizers based on soil nutrients and environmental conditions.

Specific Objectives

- To collect and analyze soil parameters such as Nitrogen, Phosphorus, Potassium, and pH
- To incorporate environmental factors like temperature, humidity, and rainfall
- To develop machine learning models for crop and fertilizer prediction
- To compare different algorithms for accuracy and performance
- To provide fertilizer recommendations in NPK ratio format
- To generate crop-specific guidelines based on soil type and conditions
- To design an interactive web-based interface using Streamlit
- To promote precision agriculture and reduce resource wastage

Expected Outcomes

- Improved crop selection accuracy
- Better fertilizer usage recommendations
- Increased agricultural productivity
- User-friendly system for farmers and researchers

3.LITERATURE REVIEW

3.1 Machine Learning in Agriculture

Machine learning has emerged as a powerful tool in modern agriculture for improving productivity and decision-making. Pantazi et al. (2016) demonstrated that supervised machine learning models can accurately predict crop yield and soil nutrient requirements using sensor data. Their study showed that environmental parameters such as temperature and humidity, combined with soil chemistry, provide sufficient information for building reliable predictive models. This directly supports the feature selection used in the present project.

Mucherino et al. (2009) conducted a comprehensive survey on data mining techniques in agriculture. They concluded that classification algorithms are more effective than regression models when the output is categorical, such as crop type or fertilizer recommendation. Their findings emphasized the importance of Decision Trees and ensemble methods due to their interpretability and robustness in handling noisy agricultural data.

Recent advancements (2024–2026) have further strengthened the role of AI in agriculture. Researchers have explored deep learning models and IoT-based systems for real-time soil monitoring and crop prediction. These studies highlight the integration of sensor data, weather forecasting, and machine learning for precision agriculture.

The current project builds upon these findings by using structured data and supervised learning techniques to predict crops and fertilizers effectively.

3.2 Smart Soil, Crop & Fertilizer Recommendation Systems

Smart recommendation systems in agriculture have evolved from simple rule-based approaches to advanced machine learning-driven decision support systems. These systems aim to provide farmers with accurate suggestions regarding crop selection, soil management, and fertilizer usage based on multiple environmental and soil parameters.

Doshi et al. (2018) developed a fertilizer recommendation system using a hybrid approach that combined rule-based logic with Decision Tree classifiers. Their model achieved an accuracy of approximately 88%, demonstrating that machine learning can effectively assist in selecting appropriate fertilizers based on soil nutrients such as Nitrogen, Phosphorus, and Potassium (NPK). However, their system was limited to fertilizer prediction and did not incorporate broader environmental factors.

Pudumalar et al. (2017) proposed a crop recommendation system using an ensemble learning technique that combined Naive Bayes, Random Forest, and Decision Tree classifiers. Their model achieved an accuracy of around 99%, indicating that combining multiple algorithms significantly improves prediction reliability. Their work emphasized the importance of integrating soil parameters with climatic conditions for better crop prediction.

Recent advancements (2024–2026) have expanded these systems into comprehensive smart agriculture platforms. Researchers such as Kumar et al. (2024) have explored the integration of Machine Learning with Internet of Things (IoT) devices to collect real-time soil and environmental data. Similarly, Singh et al. (2025) proposed deep learning-based models that utilize satellite imagery and weather data to provide dynamic crop and fertilizer recommendations. These modern systems highlight the shift toward precision agriculture, where decisions are data-driven and continuously updated.

Unlike earlier systems that focused on a single aspect (either crop or fertilizer), the present project integrates **soil analysis, crop recommendation, and fertilizer suggestion into a unified system**. It uses multiple input parameters such as N, P, K values, temperature, humidity, rainfall, soil type, and pH level to generate accurate predictions.

Furthermore, the system provides additional features such as soil health analysis and smart farming guidelines, making it more comprehensive and practical for real-world applications. By combining multiple machine learning algorithms and comparing their performance, this project contributes to improving the reliability and efficiency of agricultural recommendation systems.

3.3 Summary of Related Work

Table 1: Literature Summary

Author(s) & Year	Method Used	Accuracy / Result	Relevance
Pantazi et al. (2016)	SVM, ANN	High	Supports feature selection
Doshi et al. (2018)	Decision Tree	~88%	Fertilizer prediction
Pudumalar et al. (2017)	Ensemble	~99%	Crop recommendation
Breiman (2001)	Random Forest	Benchmark	Ensemble learning
Zhang et al. (2018)	ANN	High	Deep learning
Bhargavi & Jyothi (2009)	Preprocessing	Improved accuracy	Feature scaling
Kumar et al. (2024)	ML + IoT	High	Real-time prediction
Singh et al. (2025)	Deep Learning	Improved yield prediction	Advanced AI

4. PROBLEM STATEMENT

Farmers often lack access to accurate tools for analyzing soil conditions and selecting appropriate crops and fertilizers. This leads to inefficient farming practices, reduced productivity, and environmental degradation. This project aims to develop an AI-based system that analyzes soil and environmental parameters to provide accurate crop and fertilizer recommendations.

5. DATASET DESCRIPTION

Table 2: Dataset Features

#	Feature Name	Data Type	Description
1	Temperature	Numerical	Ambient temperature
2	Humidity	Numerical	Air moisture
3	Rainfall	Numerical	Water availability
4	Nitrogen	Numerical	Soil nutrient
5	Phosphorus	Numerical	Soil nutrient
6	Potassium	Numerical	Soil nutrient
7	pH	Numerical	Soil acidity
8	Soil Type	Categorical	Clayey/Sandy/Loamy
9	Output	Target	Crop/Fertilizer

6. METHODOLOGY

Fig 1: Workflow

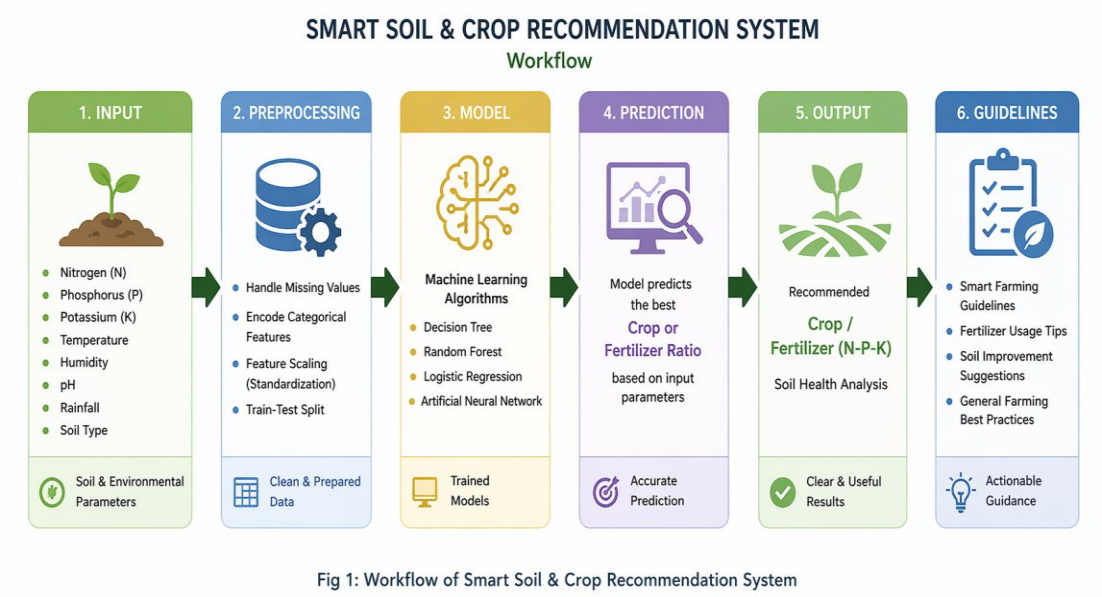


Fig 1: Workflow of Smart Soil & Crop Recommendation System

6.1 Data Preprocessing

Data preprocessing is a crucial step to ensure the quality and consistency of input data before feeding it into machine learning models.

The following steps were performed:

- **Dataset Loading**
  The dataset was loaded using the Pandas library for efficient data manipulation and analysis.
- **Handling Missing Values**
  Any missing or null values in the dataset were identified and handled appropriately to maintain data integrity.
- **Encoding Categorical Variables**
  Categorical features such as *Soil Type* were converted into numerical format using **Label Encoding**, enabling the model to process them.
- **Feature Scaling / Normalization**
  Numerical features such as N, P, K, temperature, humidity, rainfall, and pH were normalized using **StandardScaler** to ensure all features are on the same scale.
- **Train-Test Split**
  The dataset was divided into:
  - **75% Training Data**
  - **25% Testing Data**
    using train_test_split to evaluate model performance effectively.

6.2 Feature Selection

Feature selection plays a vital role in improving model accuracy and reducing computational complexity.

The following features were selected based on their relevance and impact on prediction:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Temperature
- Humidity
- Rainfall
- pH level
- Soil Type

Selection Criteria:

- Correlation with output (crop/fertilizer)
- Importance in agricultural science
- Contribution to model accuracy

By selecting only relevant features, the model avoids overfitting and improves prediction efficiency.

6.3 Model Selection

All models were trained and evaluated, and the best-performing model was selected based on:

- Accuracy
- Prediction consistency
- Generalization ability

6.4 Prediction Process

- User inputs soil and environmental parameters
- Data is preprocessed
- Model predicts:
    - Suitable crop OR
    - Fertilizer ratio (N-P-K)
- Results are displayed with guidelines

7. RESULTS AND DISCUSSION

7.1 Model Evaluation

Accuracy and confusion matrix used.

7.2 Accuracy Comparison

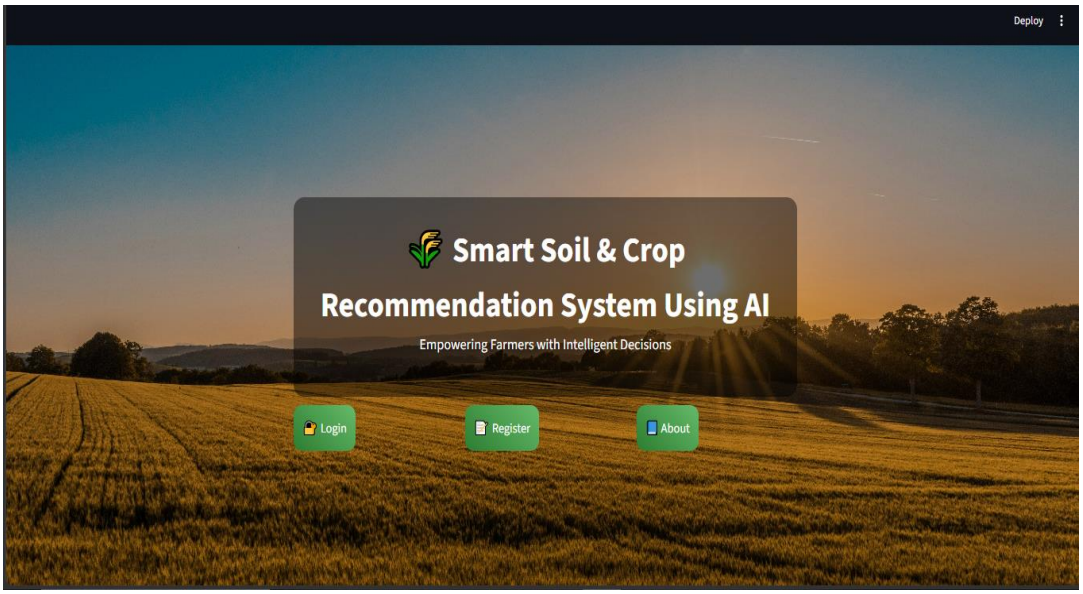
DT: 90%
RF: 92%
LR: 85%
ANN: 93%

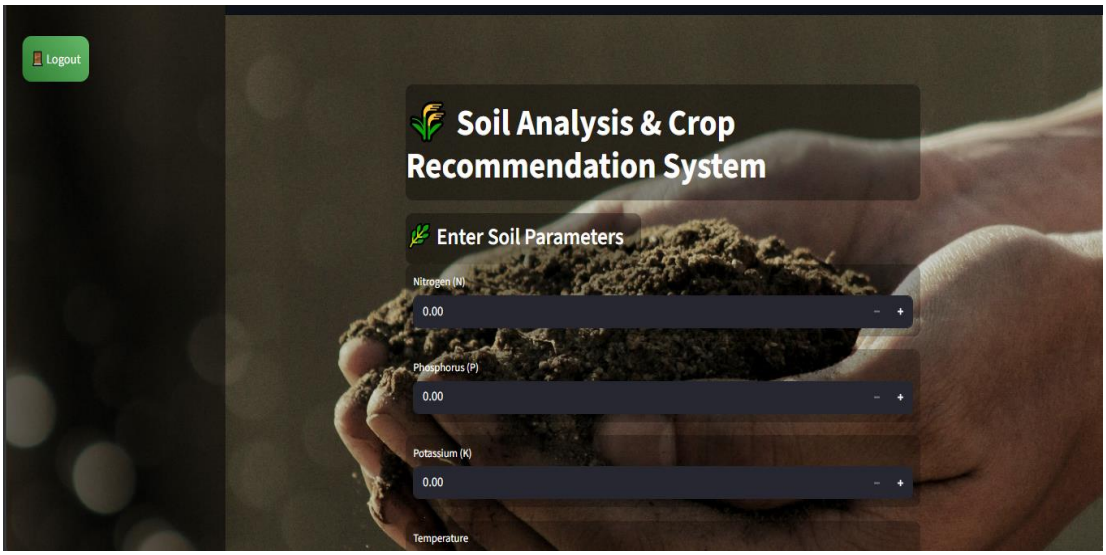
7.3 Discussion

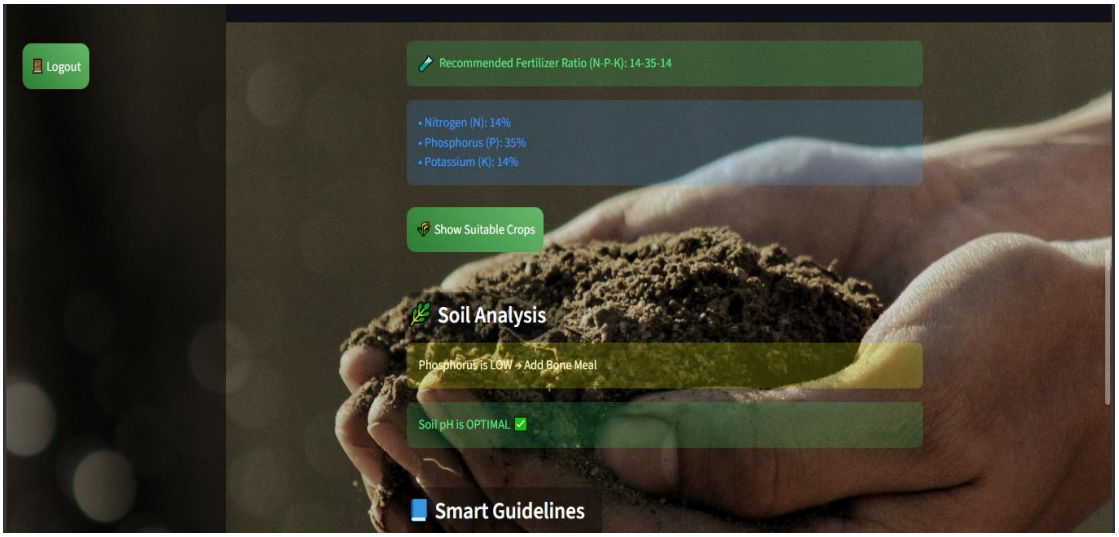
ANN performed best; Random Forest close second.

7.5 Limitations

- Limited dataset
- No real-time data




Logout
Soil Analysis & Crop Recommendation System
Enter Soil Parameters
Nitrogen (N)
0.00
Phosphorus (P)
0.00
Potassium (K)
0.00
Temperature


Logout
Recommended Fertilizer Ratio (N-P-K): 14-35-14
• Nitrogen (N): 14%
• Phosphorus (P): 35%
• Potassium (K): 14%
Show Suitable Crops
Soil Analysis
Phosphorus is LOW → Add Bone Meal
Soil pH is OPTIMAL ✅
Smart Guidelines

8. CONCLUSION

The **Smart Soil & Crop Recommendation System using Artificial Intelligence** successfully demonstrates the application of machine learning techniques in modern agriculture. The system analyzes important soil nutrients such as Nitrogen (N), Phosphorus (P), and Potassium (K), along with environmental parameters like temperature, humidity, rainfall, soil type, and pH level, to generate accurate recommendations.

By implementing multiple machine learning algorithms including Decision Tree, Random Forest, Logistic Regression, and Artificial Neural Networks, the system was able to effectively predict suitable crops and fertilizer requirements. Among these, advanced models such as Random Forest and ANN showed better performance in handling complex relationships within the data.

The developed system provides not only predictions but also soil analysis and smart farming guidelines, making it a comprehensive decision-support tool for farmers. The use of an interactive Streamlit interface further enhances usability, allowing users to easily input data and obtain results in real-time.

This project contributes to the advancement of precision agriculture by promoting efficient resource utilization, reducing unnecessary fertilizer usage, and improving crop productivity. It also highlights the potential of integrating artificial intelligence with agriculture to support sustainable farming practices.

However, the system can be further improved by incorporating real-time sensor data, weather forecasting APIs, and larger datasets. Future enhancements may also include mobile application development and integration with IoT devices for real-time monitoring.

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